

ZEBIRI RABAH

EMBEDDED SYSTEMS ENGINEER

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PROFESSIONAL SUMMARY

Embedded Systems Engineer and recent M2 graduate from the Faculty of Technology, University of Boumerdes (UMBB), specializing in low-level firmware development and hardware-software co-design. Experienced in writing bare-metal C for STM32 (ARM Cortex-M) microcontrollers, including custom peripheral drivers for SPI, I2C, and UART. Engineered a secure, collision-free LoRa relay network using AES-128 encryption and validated its performance across real-world hardware environments. Brings a foundation in RTOS, and end-to-end IoT system design.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, ARM Assembly, Python.
- **Edge AI & Machine Learning Fundamentals.**
- **Microcontrollers & Protocols:** STM32 (ARM Cortex-M), Bare-Metal & HAL Development, SPI, I2C, UART, LoRa/LoRaWAN
- **Operating Systems & Tools:** FreeRTOS, Linux Fundamentals, STM32CubeIDE, STM32CubeMX, Keil uVision, Arduino IDE, NeoVim
- **Spoken Languages:** Arabic (Fluent), English (Fluent), French (Beginner)

LATEST PROJECTS

Off-Grid Emergency System

- Engineered a cellular-independent emergency tracking and communication system designed to protect individuals and streamline rescue operations in remote, off-grid, or disaster-stricken areas.

Secure Multi Tx/Rx LoRa Relay Networks with Collision-free system “FYP thesis 19.5/20”

STM32F103C8T6 | Ra-02 (SX1278) LoRa Modules | AES-128 CBC Encryption | C/C++

- Engineered a secure, three-tier LoRa relay network and designed an algorithm to prevent collisions between multiple senders and receivers.
- Implemented AES-128 CBC encryption to secure end-to-end data transmission between nodes operating on the 433MHz band.

EDUCATION

Faculty of Technology, University of Boumerdes (UMBB)

2021 - 2026

Master's Degree in Embedded Systems Engineering

University Savoie Mont Blanc

January 2026 - February 2026

LoRa and LoRaWAN for the Internet of Things

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Electro Club — External relationships Department Member

2025 - 2026

- ER and Embedded systems member in the club, while helping the undergraduate students to sharpen their skills.

Google Developer Club — External relationships Department Member

2023 - 2024

- Speaker Outreach & Coordination — Contacted and arranged experts and professors for club lectures.